

Geofray Paul J

AI Engineer · Medical Imaging and MLOps

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SUMMARY

AI engineer for medical imaging: segmentation and classification models for computational pathology (gigapixel whole-slide images) and computational radiology (neuroimaging: MRI, CTA), and the infrastructure that trains and serves them, from nnU-Net failure analysis to containerised inference and Django/Celery services on a DGX A100. Sole author of an accepted MICCAI 2026 workshop paper; four solo MICCAI 2026 challenge entries.

EXPERIENCE

Junior AI Engineer, Sudha Gopalakrishnan Brain Centre, IIT Madras Chennai · November 2025 – present

- **CD34 microvessel segmentation.** Built an nnU-Net v2 pipeline that quantifies microvessel density across gigapixel whole-slide images (Dice 0.71); removed tile-boundary artefacts with Gaussian-weighted overlap-averaged inference.
- **Grey/white matter segmentation.** Trained HookNet on multiple sclerosis H&E whole-slide images (Dice 0.85 grey matter, 0.75 white matter). Traced a flat boundary metric to coarse white-matter annotations, not the model, with a per-class surface-Dice tolerance sweep; redirected effort to re-annotation instead of further training.
- **IHC analysis platform.** Multi-marker inference service (CD68, CD34, astrocyte) behind one React, Django and Celery/Redis interface, deployed on a DGX A100; resolved container crashes on TIFF workloads of up to 150 GB.
- **Fetal brain skull-stripping.** nnU-Net and nnSAM models for post-mortem fetal brain MRI; found and fixed a silent orientation error that transposed predictions, and set the brain-boundary convention against a public fetal brain atlas. nnU-Net trained on masks including CSF, with fold 0 fine-tuned on expert-annotated masks.
- **Specimen tracking system.** Mapped the brain-specimen lifecycle (acquisition to QR-coded tracking, scanning and storage) and its physical edge cases through interviews with lab staff; designed the relational data model and ORM layer, built the tracking interface, and delivered BPMN/UML provenance diagrams to the CSO.
- **GPU infrastructure.** Set up persistent training on a DGX A100 in Docker-isolated environments under tmux.
- **Web.** Built the institute's public website (sgbc.humanbrain.in) and BRICS 2026 Neuroscience Symposium pages solo in React.

Career break (health), June 2024 – October 2025. Fully recovered.

MICCAI 2026 CHALLENGES (SOLO ENTRIES)

- **AIMS-TBI 2026,** TBI lesion detection and segmentation (T1 MRI). nnU-Net ensemble: 8th of 22 entries on the final detection leaderboard (balanced accuracy 0.846); 11th of 17 on segmentation (lesion Dice 0.505) (official ranking announced 1 October 2026). Traced boundary error to a small-lesion recall tail: 7.2% of cases failed outright, none above median lesion volume.
- **MICCAI Educational Challenge 2026,** finalist. *Your GPU Is Lying To You:* six instruments that reported success while being wrong during a challenge submission, and a ten-point preflight checklist. Code public.
- **TopAneu 2026, Task 2,** vessel-specific intracranial aneurysm segmentation (CTA, MRA). 10th of 30 entries, Task 2 final test phase. nnU-Net detection, then gradient-boosted vessel-site classification on laterality-invariant geometry; 218 s per case against a 720 s limit. Most errors were correct detections assigned to the wrong vessel.
- **RARE26 (EndoVis),** Barrett's neoplasia classification at 1:100 prevalence. Five-model ConvNeXt ensemble for Barrett's neoplasia detection; both internal estimates overshot the platform result, the held-out-centre estimate least. Method report submitted; co-author of the organisers' challenge overview paper (in preparation).

PUBLICATION

Geofray Paul J. *Diagnosing the Recall Tail: A Class-Imbalance Study of nnU-Net for Detection and Segmentation of Heterogeneous Moderate–Severe TBI Lesions.* AIMS-TBI Workshop, MICCAI 2026. Springer LNCS, accepted, to appear.

TECHNICAL SKILLS

Medical imaging ML: PyTorch, nnU-Net v2 (ResEnc), MONAI, HookNet, timm (ConvNeXt), gradient-boosted trees

Foundation models: SAM, MedSAM, nnSAM, PathoSAM, UNI, CONCH; LoRA fine-tuning

Imaging data: gigapixel whole-slide images, DICOM, NiftI, MRI preprocessing and orientation checks, annotation quality control

Evaluation: grouped and leave-one-centre-out cross-validation, bootstrap CIs, Dice, HD95, surface Dice, AUROC

MLOps: Docker, CUDA, DGX A100, AMP and bf16, DDP, Nsight Systems, Linux, tmux; time-limited containerised inference

Software: Python, TypeScript, SQL; Django REST Framework, Celery, Redis; React, Next.js

LLM and VLM: vLLM serving on RunPod serverless (Mnemosyne); Gemini agent over a full-context knowledge base (portfolio assistant); VLM chart-reading benchmark with exact, code-generated answers (ARBITER, in progress)

EDUCATION AND LANGUAGES

B.Tech, Computer Science and Engineering (Artificial Intelligence)

2020 – 2024

Karunya Institute of Technology and Sciences, Coimbatore

Languages: English (fluent), Tamil (native), Spanish (A2).